

CoastWatch Resources for using Oceanographic Satellite Data

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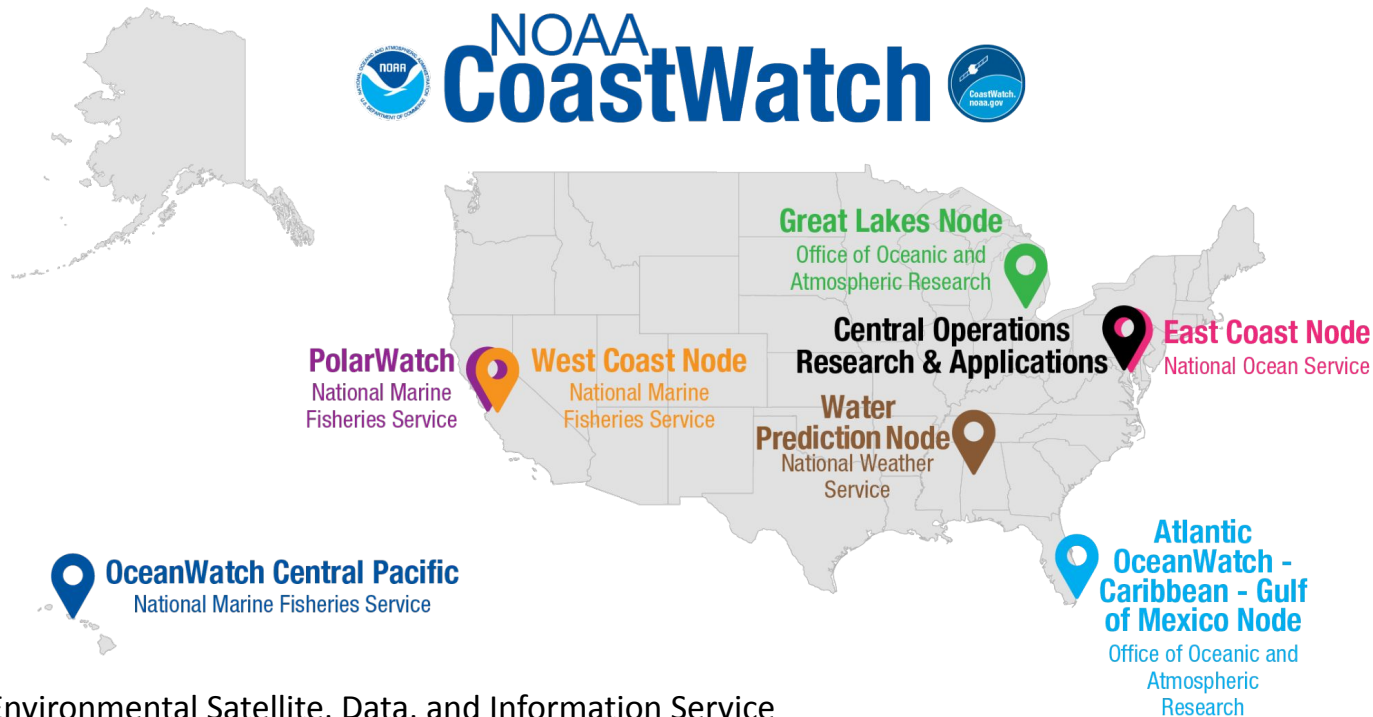
Glider Hack-a-Thon, Sept 14-18, 2026

coastwatch.info@noaa.gov



NOAA CoastWatch is a national program funded by NESDIS¹

MISSION: PROVIDE ACCESS TO AND PROMOTE THE USE OF SATELLITE DATA PRODUCTS
for oceanic, freshwater, & polar applications

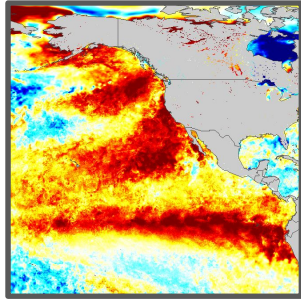


¹National Environmental Satellite, Data, and Information Service

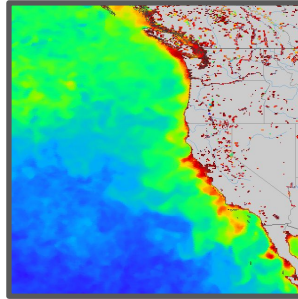


What environmental measurements can be made with satellites?

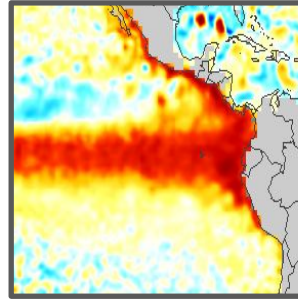
Sea Temperature



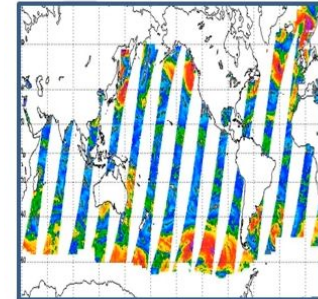
Chlorophyll a



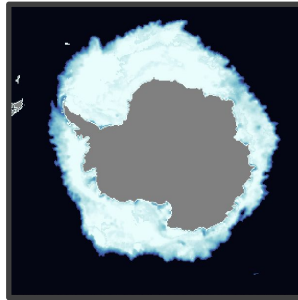
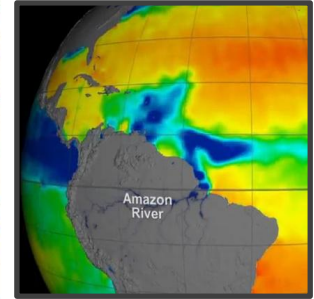
Sea Level



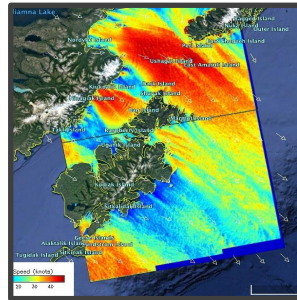
Ocean Winds



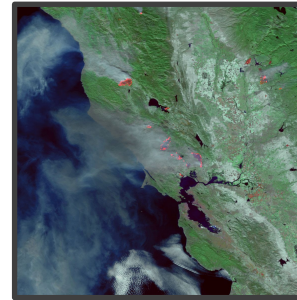
Salinity



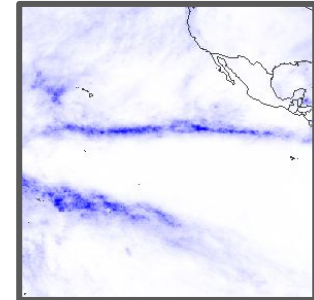
Sea Ice



**Synthetic
Aperture Radar**



Imagery



Rainfall

For how long have these measurements been made?



- Sea Ice
- Sea Surface Temperature (SST)
- Sea Surface Height (SSH)
- Chlorophyll (ocean color)
- Rainfall
- Surface Vector Winds (SWV)
- Sea Surface Salinity

For many applications we want to know how the oceans are changing over time, so we need long timeseries of consistent measurements



For how long have these measurements been made?



- Sea Ice since 1978¹
- Sea Surface Temperature (SST) since 1981
- Sea Surface Height (SSH) since 1992
- Chlorophyll (ocean color) since 1997²
- Rainfall since 1997
- Surface Vector Winds (SWV) since 1999³
- Sea Surface Salinity since 2011

For many applications we want to know how the oceans are changing over time, so we need long timeseries of consistent measurements

¹The *consistently processed* satellite passive microwave record of sea ice concentration begins in 1978, but other data extends to 1966.

²The continuous record of ocean color sensors extends back to 1997, but the CZCS mission flew from 1979-1986.

³Wind speed, without direction, dates back to 1988



CoastWatch Training Website

<https://coastwatch-training.github.io/>



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CoastWatch Training Website

This site provides information on CoastWatch Satellite Training courses such as upcoming or past trainings, training tutorials, course lectures, and more to support scientists working with oceanographic satellite data.

What you'll find here

- Step-by-step coding and software tutorials for accessing and analyzing satellite data
- Training materials developed through NOAA CoastWatch courses, including information on upcoming and past trainings
- Resources and support options, including scheduled office hours, to help you integrate satellite data into research and management workflows

Not sure where to start?

• Interested in participating in a training course?

Visit the [Training Classes](#) page to see upcoming trainings or past trainings to get an idea of what to expect.

• Looking for hands-on examples?

Check out the [Training Tutorials](#) to explore workflows in Python, R, and MATLAB, as well as software tutorials exploring NASA's Panoply and CoastWatch Utilities.

• Interested in reviewing lecture content at your own pace?

Browse our collection of lectures and presentations on the [Course Lectures](#) page. If you want course specific content, browse our [training class pages](#) for course specific presentations and materials.

• Want a preview of participant projects?

Check out [archived presentation slides](#) developed by past attendees showcasing how they applied course concepts to their own data and research.

• Need help or have questions?

The [Help](#) page points you to FAQs, resources, and support options.



CoastWatch Training Website Course Lectures

<https://coastwatch-training.github.io/>

- More than 20 presentations on various aspects of satellite data
- Presentations kept to ~ 20 minutes
- Can be viewed online or downloaded
- Video and transcripts available for most presentations



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Click the **eye** icon to preview the lecture slides directly on this page, **download** to go to the source file, **play button** to view the narrated lecture, and **clipboard** to read the transcript of the narrated lecture.

Category All 22 shown

Lecture	Preview	Download	Video	Transcript
Introduction to CoastWatch An overview of the NOAA/NESDIS CoastWatch program, it's different regional nodes, and the satellite products and services it offers.			—	—
Introduction to ERDDAP This presentation introduces ERDDAP and demonstrates how it simplifies discovering, visualizing, subsetting, and downloading gridded and tabular oceanographic datasets from multiple satellite and in situ data servers through a consistent web interface and URL-based queries.				
Satellite 101, Part 1 This presentation covers the basics of ocean remote sensing, part 1.				
Satellite 101, Part 2 This presentation covers EMR (electromagnetic radiation), atmospheric correction, and the differences between passive and active sensors.				
Tools and Strategy This presentation provides an overview of CoastWatch data services and tools, showing how to find, visualize, and access satellite oceanographic data using NetCDF files, data viewers, and ERDDAP to simplify working with multiple data servers and formats.				
How to Select the Best Dataset for Your Project This presentation provides a practical framework for choosing the most appropriate satellite dataset for a project by evaluating tradeoffs in temporal and spatial coverage, resolution, latency versus quality, and tolerance for missing data, with examples drawn from common CoastWatch products.				
Intro to Using R with Satellite Data This presentation introduces how to use R with ERDDAP, focusing on the rerddapXtracto package to extract, visualize, and analyze satellite oceanographic data for applications such as track matchups, polygon-based regions, time series, and habitat analyses using reproducible R workflows.				



CoastWatch Training Website: Participant Slides

<https://coastwatch-training.github.io/>

- We ask participants in our full week-long courses to contribute a slide showing how they are using satellite data
- We have a curated collection of some of these slides on the training website



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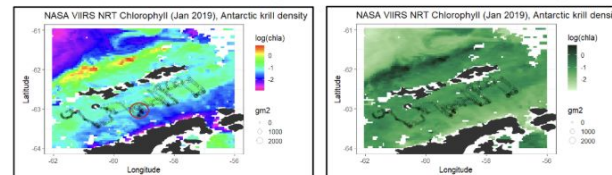
Help

Examples of Participant Slides

As part of our full, week-long courses we ask participants to complete a final slide summarizing what they were able to accomplish during the course. Each slide includes the participant's name and affiliation, a description of their project or objective, and a figure created using satellite data during the course. Participants also document the datasets and software used, and may include any challenges encountered or desired data that were not available.

Here we showcase some slides that are representative of the broad usages and applications.

Jen Walsh (NOAA/AERD): Final Project, CoastWatch Satellite Course May 2021



Chlorophyll a concentration (log) around the northern Antarctic Peninsula in January 2019 (monthly composite from NASA VIIRS Near-Real-Time, 4 km spatial resolution). Circles in the Bransfield Strait (area between the peninsula and the South Shetland Islands to the north) are Antarctic krill density values (g m^{-2}) acoustically estimated from an autonomous underwater glider deployed from December 2018 – March 2019. Left plot is a rainbow palette, right plot is the “algae” palette from the cmocean package in R. Red circle on left plot highlights area of high krill density in a low chlorophyll area.

Glider (~8 ft)



Krill (~40 mm)

IMAGE 18 OF 32



CoastWatch Training Website

Software Tutorials

<https://coastwatch-training.github.io/>

- Tutorials available on using ERDDAP, Panoply (a netCDF viewer) and CoastWatch Utilities.
- Also a good guide to troubleshooting potential errors encountered when using the R package rerddapXtracto



[Training Tutorials](#) > [Software Tutorials](#) > [Software Code Gallery](#)

Software Code Gallery

This gallery brings together hands-on, software-based tutorials for working with CoastWatch and ERDDAP data. Each tutorial focuses on a specific task—such as discovering datasets, building ERDDAP URLs, visualizing gridded and tabular data, or troubleshooting common workflows—using tools like Python, R, and related utilities.

Click the **info** icon to learn more about the software and tools used in the tutorial, **eye** icon to preview outputs, **box** to view tutorial resources, or **download** to open the source file on GitHub.

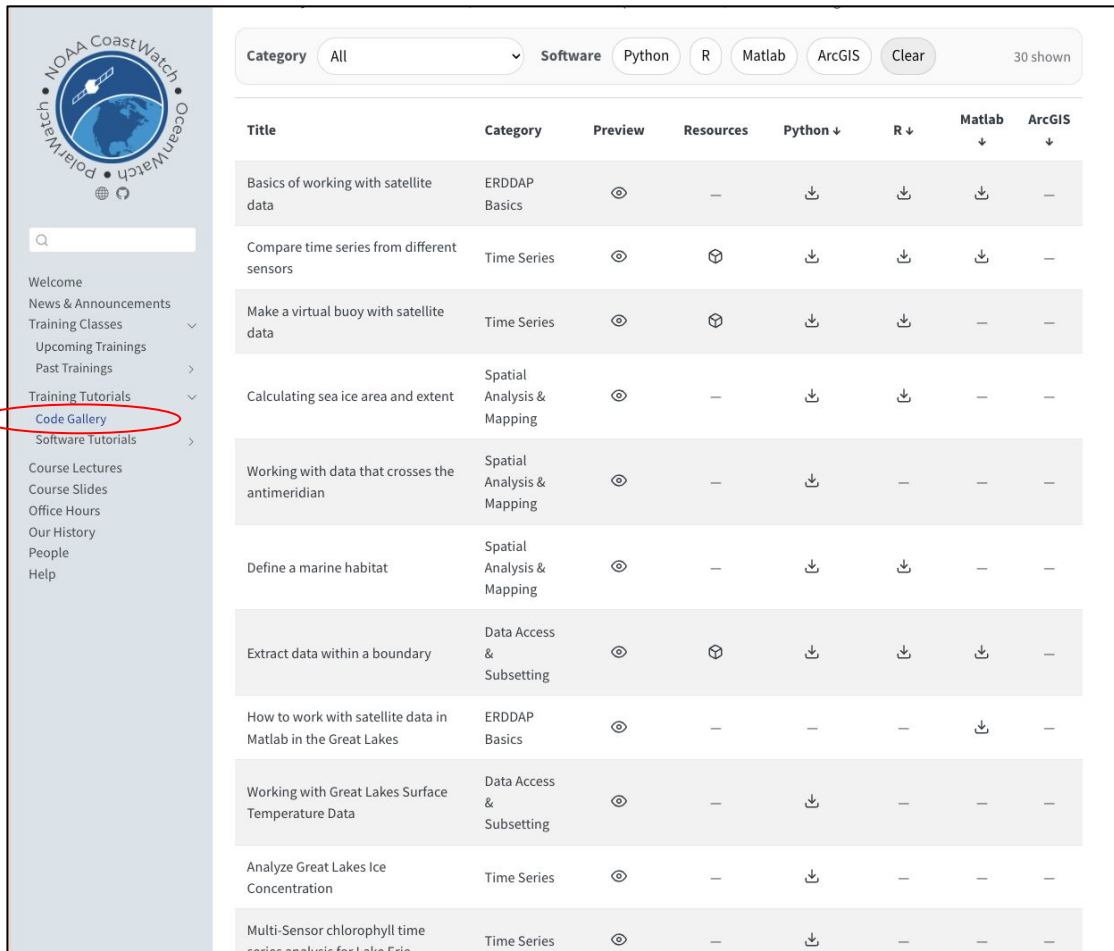
Title	Info	Preview	Resources	Download ↓
ERDDAP Basics - Using the ERDDAP data catalog	📄	👁	—	↓
ERDDAP Basics - Visualize and download data	📄	👁	—	↓
ERDDAP Basics - Understanding the ERDDAP URL	📄	👁	—	↓
ERDDAP Basics - Creating a Hovmoller plot	📄	👁	—	↓
ERDDAP Basics - Work with wind vectors	📄	👁	—	↓
ERDDAP Basics - Using tabular data	📄	👁	—	↓
ERDDAP Basics - Additional Resources	📄	👁	—	↓
Setting up a Python environment	—	👁	📦	↓
Troubleshooting rerddapXtracto	—	👁	—	↓
NetCDF and Panoply Tutorial	📄	👁	—	↓
CoastWatch Utilities	📄	👁	—	—



CoastWatch Training Website: Code Gallery

<https://coastwatch-training.github.io/>

- We have a code Gallery with notebooks demonstrating how to do example workflows.
- Code is available in R and python, some examples also have Matlab or ARCGIS versions.
- Preview option shows images produced by the code



The screenshot shows the NOAA CoastWatch Training Website's Code Gallery. The left sidebar contains a navigation menu with the following items: Welcome, News & Announcements, Training Classes, Upcoming Trainings, Past Trainings, Training Tutorials, **Code Gallery** (highlighted with a red circle), Software Tutorials, Course Lectures, Course Slides, Office Hours, Our History, People, and Help. The main content area features a table of training materials. At the top of the table, there is a 'Category' dropdown menu set to 'All', and buttons for 'Software', 'Python', 'R', 'Matlab', 'ArcGIS', and 'Clear'. A '30 shown' indicator is also present. The table has the following columns: Title, Category, Preview, Resources, Python, R, Matlab, and ArcGIS. The table lists 12 items, each with a preview icon (an eye) and download icons for various software packages.

Title	Category	Preview	Resources	Python	R	Matlab	ArcGIS
Basics of working with satellite data	ERDDAP Basics		—				—
Compare time series from different sensors	Time Series						—
Make a virtual buoy with satellite data	Time Series					—	—
Calculating sea ice area and extent	Spatial Analysis & Mapping		—			—	—
Working with data that crosses the antimeridian	Spatial Analysis & Mapping		—		—	—	—
Define a marine habitat	Spatial Analysis & Mapping		—			—	—
Extract data within a boundary	Data Access & Subsetting						—
How to work with satellite data in Matlab in the Great Lakes	ERDDAP Basics		—	—	—		—
Working with Great Lakes Surface Temperature Data	Data Access & Subsetting		—		—	—	—
Analyze Great Lakes Ice Concentration	Time Series		—		—	—	—
Multi-Sensor chlorophyll time series analysis for Lake Erie	Time Series		—		—	—	—



CoastWatch Training Website: Code Gallery

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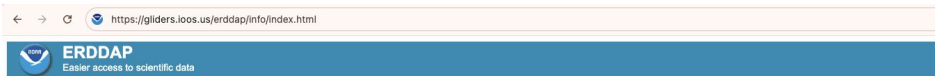
The screenshot displays the NOAA CoastWatch Training Website Code Gallery. At the top, there's a navigation bar with tabs for 'Category', 'All', 'Software', 'Python', 'R', 'Matlab', 'ArcGIS', and 'Clear'. Below this, a modal window titled 'Comparing Glider Observations with Satellite Data' is open. The modal contains a description: 'Learn how to retrieve glider and satellite data from ERDDAP, match observations in space and time, and compare chlorophyll-a, sea surface temperature, and sea surface salinity.' It features two side-by-side plots under the heading 'Python output' and 'R output'. Both plots show 'Sea Surface Temperature Comparison: CalCOFI Line 90' with 'Glider (10 M)' and 'Satellite (SST)' data. The plots show temperature (°C) on the y-axis (ranging from 12.0 to 17.5) and time on the x-axis (from 2004-01-01 to 2004-01-01). Below the plots, there's a description: 'Comparison of in situ glider temperature measurements at 10 m depth and satellite-derived sea surface temperature (SST) along CalCOFI Line 90. Points show individual observations through time, with lines included to highlight temporal patterns.' There are buttons for 'View Python tutorial', 'View R tutorial', and 'Close'. At the bottom of the modal, there's a table with columns for 'Matlab in the Great Lakes', 'Basics', 'Data Access & Subsetting', 'Time Series', and 'Time Series'. The table lists various topics like 'Working with Great Lakes Surface Temperature Data', 'Analyze Great Lakes Ice Concentration', and 'Multi-Sensor chlorophyll time series analysis for Lake Erie'.

Matlab in the Great Lakes	Basics				
Working with Great Lakes Surface Temperature Data	Data Access & Subsetting	👁	—	↓	—
Analyze Great Lakes Ice Concentration	Time Series	👁	—	↓	—
Multi-Sensor chlorophyll time series analysis for Lake Erie	Time Series	👁	—	↓	—



Glider Data ERDDAPs

1. IOOS Glider DAAC <https://gliders.ioos.us/erddap>

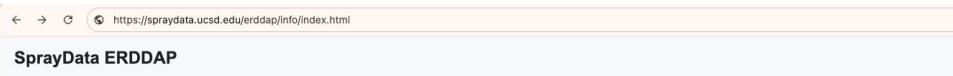


ERDDAP > List of All Datasets

2541 matching datasets, listed in alphabetical order. View page: 1 (current) 2 3

Grid DAP	Sub-set	Table Data	Make A Graph	W M S	Source Data Files	Title	Summary	FGDC, ISO, Metadata	Background Info	RSS	E mail	Institution	Dataset ID
set	data	graph	files			* The List of All Active Datasets in this ERDDAP *		I M	background			Set-net Institution	allDatasets
set	data	graph	files			amella-20180501T0000		F I M	background			Virginia Institut...	amella-20180501T0000
set	data	graph	files			amella-20200825T1929		F I M	background			Virginia Institut...	amella-20200825T1929
set	data	graph	files			amella-20201015T1436		F I M	background			Virginia Institut...	amella-20201015T1436
set	data	graph	files			amir01-20181216T0641-delayed		F I M	background			NOAA SWFSC Antarc...	amir01-20181216T0641-delayed
set	data	graph	files			amir01-20191206T0452-delayed		F I M	background			NOAA SWFSC Antarc...	amir01-20191206T0452-delayed
set	data	graph	files			amir02-20181211T1233-delayed		F I M	background			NOAA SWFSC Antarc...	amir02-20181211T1233-delayed
set	data	graph	files			amir02-20191206T1236-delayed		F I M	background			NOAA SWFSC Antarc...	amir02-20191206T1236-delayed
set	data	graph	files			amir03-20191206T0529-delayed		F I M	background			NOAA SWFSC Antarc...	amir03-20191206T0529-delayed
set	data	graph	files			angus-20190522T0000		F I M	background			Skidaway Institut...	angus-20190522T0000
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set	data	graph	files			angus-20201115T0000		F I M	background			Skidaway Institut...	angus-20201115T0000
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set	data	graph	files			angus-20211110T0000		F I M	background			Skidaway Institut...	angus-20211110T0000
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set	data	graph	files			angus-20220404T0000-delayed		F I M	background			Skidaway Institut...	angus-20220404T0000-delayed
set	data	graph	files			angus-20230122T0000		F I M	background			Skidaway Institut...	angus-20230122T0000
set	data	graph	files			angus-20230528T0051		F I M	background			Skidaway Institut...	angus-20230528T0051
set	data	graph	files			angus-20240111T0000		F I M	background			Skidaway Institut...	angus-20240111T0000
set	data	graph	files			angus-20240215T0000		F I M	background			Skidaway Institut...	angus-20240215T0000
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set	data	graph	files			bass-20150706T151619Z		F I M	background			USF	bass-20150706T151619Z
set	data	graph	files			bass-20150827T1909		F I M	background			USF	bass-20150827T1909
set	data	graph	files			bass-20160909T1733		F I M	background			University of Sou...	bass-20160909T1733
set	data	graph	files			bass-20180301T0000		F I M	background			University of Sou...	bass-20180301T0000
set	data	graph	files			bass-20180808T0000		F I M	background			University of Sou...	bass-20180808T0000
set	data	graph	files			bass-20200203T0000		F I M	background			University of Sou...	bass-20200203T0000
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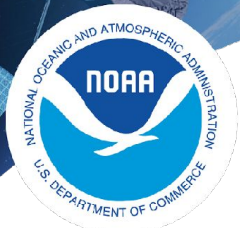
2. SprayData ERDDAP, UCSD <https://spraydata.ucsd.edu/erddap>



ERDDAP > List of All Datasets

30 matching datasets, listed in alphabetical order.

Grid DAP	Sub-set	Table Data	Make A Graph	W M S	Source Data Files	Title	Summary	FGDC, ISO, Metadata	Background Info	RSS	E mail	Institution	Dataset ID
set	data	graph	files			* The List of All Active Datasets in this ERDDAP *		I M	background			Scripps Institut...	
set	data	graph	files			California Underwater Glider Network - Along Shore Transect		F I M	background			Scripps Institut...	
set	data	graph	files			California Underwater Glider Network - Line 56		F I M	background			University of Cal...	
set	data	graph	files			California Underwater Glider Network - Line 66		F I M	background			University of Cal...	
set	data	graph	files			California Underwater Glider Network - Line 90		F I M	background			University of Cal...	
set	data	graph	files			California Underwater Glider Network - Line 93		F I M	background			University of Cal...	
data	data	graph	files			CUGN Climatology v4, Along-shore Line, Annual Anomaly, Rho Layers		I M	background			Scripps Institut...	
data	data	graph	files			CUGN Climatology v4, Along-shore Line, Mean Annual Cycle, Rho Layers		I M	background			Scripps	
data	data	graph	files			ESA OCCCI V6, 2006-2025, CUGN Region		F I M	background			Plymouth Marine L...	
set	data	graph	files			Glider observations in the Kuroshio and South China Sea		F I M	background			University of Cal...	
set	data	graph	files			Glider observations of circulation near abrupt topography		F I M	background			University of Cal...	
set	data	graph	files			Glidars in the Gulf of Mexico		F I M	background			University of Cal...	
set	data	graph	files			Glidars in the Gulf Stream		F I M	background			Woods Hole Ocean...	
data	data	graph	files			Gulf Stream annual cycle from Spray underwater glider measurements, 2015-2023, Daily		F I M	background			Woods Hole Ocean...	
data	data	graph	files			Gulf Stream mean from Spray underwater glider measurements, 2015-2023		F I M	background			Woods Hole Ocean...	
set	data	graph	files			NRT Data from Spray Glider Mission (sp036-20251119T1927)		F I M	background			Scripps	
set	data	graph	files			Observations of the Almeria-Oran front in the Mediterranean Sea using Spray underwater gliders		F I M	background			University of Cal...	
set	data	graph	files			Origins of the Kuroshio and Mindanao Current		F I M	background			University of Cal...	
set	data	graph	files			Repeat Observations by Gliders in the Equatorial Region		F I M	background			University of Cal...	
set	data	graph	files			Seasonal-to-Interannual Modulation of Fine-Scale Thermohaline Structure		F I M	background			University of Cal...	
set	data	graph	files			Solomon Sea Ocean Transport from Gliders		F I M	background			University of Cal...	
set	data	graph	files			Southern California Water Temperature Index, 10-day, v1.0		I M	background			Scripps Institut...	
set	data	graph	files			Southern California Water Temperature Index, Monthly, v1.0		I M	background			Scripps Institut...	
set	data	graph	files			Spray glider data in support of mooring observations from the CORC project in the California Current starting 2007		F I M	background			University of Cal...	
set	data	graph	files			Spray Glider Data, Science Quality, Global, High-Resolution, 2005-Present		F I M	background			Scripps Institut...	
set	data	graph	files			Spray glider observations in support of MAB-SWOT		F I M	background			Woods Hole Ocean...	
set	data	graph	files			Spray glider observations in support of NASCAR		F I M	background			University of Cal...	
set	data	graph	files			Spray glider observations in support of PEACH		F I M	background			University of Cal...	
set	data	graph	files			(DEPRECATED) Southern California Water Temperature Index v0.2 DEPRECATED		I M	background			University of Cal...	



Introduction to ERDDAP

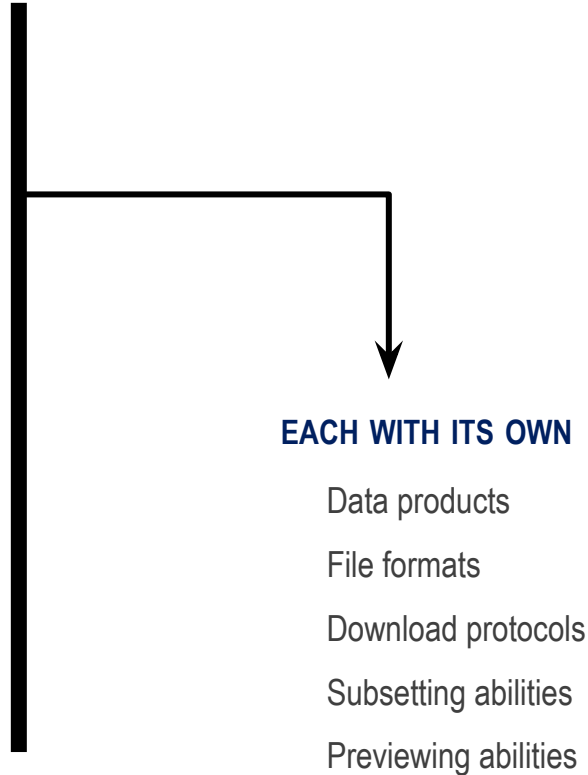
NOAA CoastWatch Satellite Course



Accessing satellite data can be challenging

A SHORT LIST OF DATA SERVERS

NOAA CoastWatch Central Operations
NOAA Center for Satellite Applications and Res.
NOAA Office of Satellite and Products
NOAA National Centers for Environmental Info.
NOAA Comprehensive Large Array-data
Stewardship System (CLASS)
NASA Jet Propulsion Laboratory PO.DAAC
NASA Ocean Biology (OB.DAAC)
NASA Goddard Space Flight Center
European Space Agency
EUMETSAT
Japan Aerospace Exploration Agency



ERDDAP¹ – designed to make data access easier

DATA AGGREGATION

LOCAL STORAGE

Internal Servers
Database
RAID

REMOTE SERVERS

NSIDC
NCEI
JPL PO.DAAC
NESDIS STAR

DATA DISTRIBUTION

Automated Scripts
Web-Based Applications
Download By Hand
Software Applications



¹ERDDAP was developed at NOAA/NMFS/SWFSC/ERD by Bob Simons

ERDDAP provides a simple, consistent way to:

- Subset datasets temporally and spatially
- Distribute both gridded and non-gridded (tabular) data
- Download data in > 30 formats
- Data requests defined within URLs, allowing:
 - Access data within analysis tools (R, Matlab, python)
 - Machine-to-machine data exchange

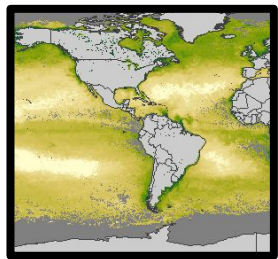
Over 85 ERDDAPs exist worldwide

Over a dozen different ERDDAPs in NOAA

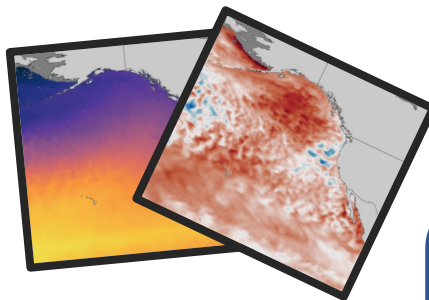
**ERDDAP is one of the recommended data servers
in NOAA's Data Access Procedural Directive**

Search for data across multiple ERDDAPs at erddap.com

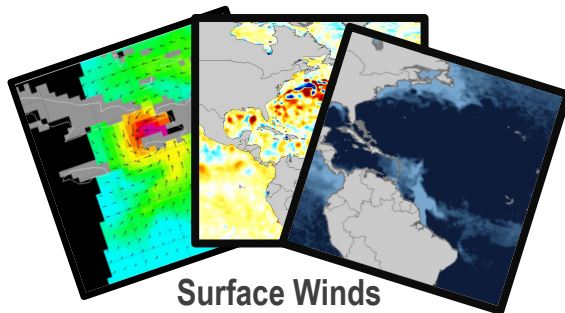
NOAA/ERD ERDDAP contains > 1000 satellite datasets



**Chlorophyll
Primary Productivity**



**SST
SST Anomaly**



**Surface Winds
Sea Surface Salinity
Sea Surface Height and Anomaly**

0.5 – 1 million data requests per day

- Daily, weekly, and monthly composites
- Blended products
- Interpolated products (gap free)
- All level 3 or 4 products (i.e. on a regular XY grid)

This ERDDAP is maintained jointly by the [SWFSC Environmental Research Division](#) and the [West Coast Node\(WCN\)](#) of NOAA's CoastWatch program



ERD ERDDAP data catalog has >400 non-satellite datasets

In Situ Measurements

- Animal Telemetry Network
- ARGO floats
- TAO/TRITON, RAMA, & PIRATA Buoys
- IOOS In Situ Sensors
- Glider Data
- Global Temperature and Salinity Profile Programme
- HF Radar Currents
- GLOBEC Northeast Pacific
- NOAA CO-OPS Sensors
- NDBC buoys

Field Sampling

- CalCOFI
- California Fish Landings
- Farallon Island Seabirds
- NWFSC Habitat Use
- SWFSC Rockfish

Underway Data

- NOAA Vessels
- UNOLS Vessels

Models, Climatologies

- OSCAR Sea Surface Velocity
- SODA Model

Models, Climatologies (cont.)

- NOAA Coastal Relief Model
- NOAA RTOFS Forecast Model
- NOAA RTOFS Nowcast Model
- NOAA World Ocean Atlas
- NOAA Seafloor Topography
- SWFSC Upwelling Index
- Navy NAVGEM Model
- Navy NOGAPS Model
- NCEP/NCAR Reanalysis
- USGS Topography
- NASA/NOAA CCMP Wind Atlas
- Navy HYCOM Model
- Navy FNMOC Forecast Model



The ERDDAP interface is functionally beautiful



ERDDAP

Easier access to scientific data

ERDDAP > List of All Datasets

1392 matching datasets, listed in alphabetical order. View page: 1 (current) 2 .

Grid DAP Data	Sub-set	Table DAP Data	Make A Graph	W M S	Source Data Files	Access-ible	Title	Sum-mary	FGDC, ISO, Metadata	Back-ground Info	RSS	E mail	Institution	Dataset ID
	set	data	graph			public	* The List of All Active Datasets in this ERDDAP *		M	background			NOAA NMFS SWFSC E...	allDatasets
data			graph			public	AMSRE Model Output, obs4MIPs NASA-JPL, Global, 1 Degree, 2002-2010, Monthly		F I M	background			Remote Sensing Sy...	jplAmsreSstMon
data			graph	M		public	AMSRE Model Output, obs4MIPs NASA-JPL, Global, 1 Degree, 2002-2010, Monthly, Lon+/-180		F I M	background			Remote Sensing Sy...	jplAmsreSstMon_LonPM180
		data	graph		files	public	AN EXPERIMENTAL DATASET: Underway Sea Surface Temperature and Salinity Aboard the Oleaner, 2007-2010		F I M	background			NOAA OAR AOML	nodcPJJU
	set	data	graph			public	Animal Telemetry Network (ATN)		F I M	background			Animal Telemetry ...	gtoppAT
data			graph	M		public	Aquarius Sea Surface Salinity, L3 SMI, Version 5, 1.0°, Global, 2011-2015, 3-Month		F I M	background			NASA/GSFC OBPG	jplAquariusSSS3MonthV5
data			graph	M		public	Aquarius Sea Surface Salinity, L3 SMI, Version 5, 1.0°, Global, 2011-2015, 7-Day		F I M	background			NASA/GSFC OBPG	jplAquariusSSS7DayV5
data			graph	M		public	Aquarius Sea Surface Salinity, L3 SMI, Version 5, 1.0°, Global, 2011-2015, Daily		F I M	background			NASA/GSFC OBPG	jplAquariusSSSDailyV5
data			graph	M		public	Aquarius Sea Surface Salinity, L3 SMI, Version 5, 1.0°, Global, 2011-2015, Monthly		F I M	background			NASA/GSFC OBPG	jplAquariusSSSMonthlyV5
data			graph		files	public	Audio data from a local source.		M	background			???	testGridWav
	set	data	graph		files	public	Audio data from a local source.		M	background			???	testTableWav
data			graph	M		public	AVHRR Pathfinder Version 5.3 L3-Collated (L3C) SST, Global, 0.0417°, 1981-present, Daytime (1 Day Composite)		F I M	background			NCEI	nceiPH53std1day
data			graph	M		public	AVHRR Pathfinder Version 5.3 L3-Collated (L3C) SST, Global, 0.0417°, 1981-present, Nighttime (1 Day Composite)		F I M	background			NCEI	nceiPH53stdn1day
data			graph			public	AVISO Model Output, obs4MIPs NASA-JPL, Global, 1 Degree, 1992-2010, Monthly		F I M	background			Centre National d...	jplAvisoSshMon
data			graph	M		public	AVISO Model Output, obs4MIPs NASA-JPL, Global, 1 Degree, 1992-2010, Monthly, Lon+/-180		F I M	background			Centre National d...	jplAvisoSshMon_LonPM180
data			graph	M	files	public	C-HARM 1-Day Advanced Forecast: Pseudo-Nitzschia, cellular domoic acid, and particulate domoic acid probability, California and Southern Oregon coast		F I M	background			UCSC, UCSD	charmForecast1day
data			graph	M	files	public	C-HARM 2-Day Advanced Forecast: Pseudo-Nitzschia, cellular domoic acid, and particulate domoic acid probability, California and Southern Oregon coast		F I M	background			UCSC, UCSD	charmForecast2day
data			graph	M	files	public	C-HARM 3-Day Advanced Forecast: Pseudo-Nitzschia, cellular domoic acid, and particulate domoic acid probability, California and Southern Oregon coast		F I M	background			UCSC, UCSD	charmForecast3day
data			graph	M	files	public	C-HARM Nowcast: Pseudo-Nitzschia, cellular domoic acid, and particulate domoic acid probability, California and Southern Oregon coast		F I M	background			UCSC, UCSD	charmForecast0day
	set	data	graph			public	CalCOFI Continuous Underway Fish-Egg Sampler		F I M	background			NOAA SWFSC	erdCalCOFicufes
	set	data	graph			public	CalCOFI Cruises		M	background			NOAA SWFSC	erdCalCOFicruises

Online interface to create custom graphs

Graph Type:

- Maps (surface)
- Time-series (lines)
- Hovmöller (surface)
- Vectors (vectors)

Color:

- Choose variable in dataset

Scale:

- Choose linear or log

Color Bar:

- Choose from > 40 color palettes

File Type:

- Choose from > 40 file formats
(data and graphics)

 **ERDDAP**
Easier access to scientific data

ERDDAP > griddap > Make A Graph

Dataset Title: NOAA Coral Reef Watch Operational Daily Near-Real-Time Global 5-km Satellite Coral Bleaching Monitoring Products [✉](#) [RSS](#)

Institution: National Oceanic and Atmospheric Administration (NOAA) (Dataset ID: NOAA_DHW)

Information: [Summary](#) | [License](#) | [FGDC](#) | [ISO 19115](#) | [Metadata](#) | [Background](#) | [Data Access Form](#)

Graph Type: surface

X Axis: longitude

Y Axis: latitude

Color: CRW_SST

Dimensions

time (UTC) specify just 1 value → 2020-02-10T12:00:00Z

latitude (degrees_north) 89.975 -89.975

longitude (degrees_east) -179.975 179.975

Graph Settings

Color Bar: Continuity: Scale: N Sections:

Minimum: Maximum:

Draw land mask: Ascending

Y Axis Minimum: Maximum:

Redraw the Graph (Please be patient. It may take a while to get the data.)

Optional:

Then set the File Type: .htmlTable (File Type information)

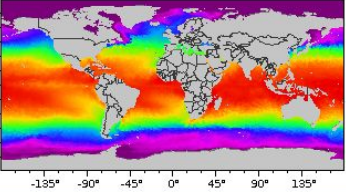
and Download the Data or an Image

or view the URL: https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW.htmlTable?CRW_

(Documentation / Bypass this form)

Click on the map to specify a new center point.

Zoom: Data Out 8x Out 2x Out In In 2x In 8x



sea surface temperature (Celsius)

NOAA Coral Reef Watch Operational Daily Near-Real-Time Global 5-km Satellite Coral Bleaching Monitoring Products (2020-02-10T12:00:00Z)

Data courtesy of National Oceanic and Atmospheric Administration (NOAA)

Online interface to download data

Ca
lor
Fil

.asc - View OPeNDAP-style ISO-8859-1 comma-separated
.csv - Download a ISO-8859-1 comma-separated
.csvp - Download a ISO-8859-1 .csv file with line
.csv0 - Download a ISO-8859-1 .csv file without c
.das - View the dataset's metadata via an ISO-885
.dds - View the dataset's structure via an ISO-885
.dods - OPeNDAP clients use this to download the
.esriAscii - Download an ISO-8859-1 ESRI ASCII fi
.fgdc - View the dataset's UTF-8 FGDC .xml metad
.graph - View a Make A Graph web page.
.help - View a web page with a description of grid
.html - View an OPeNDAP-style HTML Data Acces
.htmlTable - View a UTF-8 .html web page with th
.iso19115 - View the dataset's ISO 19115-2/19139
.itx - Download an ISO-8859-1 Igor Text File. Each
.json - View a table-like UTF-8 JSON file (missing
.jsonlCSV1 - View a UTF-8 JSON Lines CSV file wi
.jsonlCSV - View a UTF-8 JSON Lines CSV file wit
.jsonlKVP - View a UTF-8 JSON Lines file with Key
.mat - Download a MATLAB binary file.
.nc - Download a NetCDF-3 binary file with COARI
.ncHeader - View the UTF-8 header (the metadata
.ncml - View the dataset's structure and metadata
.nccsv - Download a NetCDF-3-like 7-bit ASCII NC
.nccsvMetadata - View the dataset's metadata as
.ncoJson - Download a UTF-8 NCO lvl=2 JSON fil
.odvTxt - Download time,lat,lon,otherVariables as
.timeGaps - View a UTF-8 list of gaps in the time
.tsv - Download a ISO-8859-1 tab-separated text
.tsvp - Download a ISO-8859-1 .tsv file with line 1

 **ERDDAP**
Easier access to scientific data

ERDDAP > griddap > Data Access Form

Dataset Title: SST and SST Anomaly, NOAA Global Coral Bleaching Monitoring, 5km, V.3.1, Monthly, 1985-Present  

Institution: NOAA/NESDIS/STAR Coral Reef Watch program (Dataset ID: NOAA_DHW_monthly)

Information: [Summary](#) | [License](#) | [FGDC](#) | [ISO 19115](#) | [Metadata](#) | [Background](#) | [Files](#) | [Make a graph](#)

Dimensions	Start	Stride	Stop	Size	Spacing
☒ time (UTC)	1985-01-16T00:00:00Z	1	2020-11-16T00:00:00Z	431	30 days 10h 29m 35s (uneven)
☒ latitude (degrees_north)	60.025	1	60.025	3600	-0.05 (uneven)
☒ longitude (degrees_east)	-169.975	1	-169.975	7200	0.05 (uneven)

Grid Variables (which always also download all of the dimension variables)

☐ sea_surface_temperature (degree_C)

☐ mask (Pixel characteristics flag array, pixel_classification)

☒ sea_surface_temperature_anomaly (degree_C)

File type: [\(more info\)](#)



[\(Documentation / Bypass this form\)](#)

Submit (Please be patient. It may take a while to get the data.)



Deconstructing an ERDDAP data request URL

NOAA_DHW_monthly.largePng?sea_surface_temperature[(2019-09-21T12:00:00Z)]

Example of a URL data request

Base URL: <https://coastwatch.pfeg.noaa.gov/erddap/griddap/>

Dataset ID: NOAA_DHW_monthly

File Type: .largePng (.nc, .mat, .json, .geotif, .kml, .csv...)

Data Request Begins ?

Variable: sea_surface_temperature

Time range: [(2019-01-15T12:00:00Z):(2019-01-15T12:00:00Z)]

Latitude Range: [(70):(-10)]

Longitude Range: [(-180):(-100)]

[(70):(-10)][(-180):(-100)]

[https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea_surface_temperature\[\(2019-01-15T23:00:00Z\)\]\[\(70\):\(-10\)\]\[\(-180\):\(-100\)\]](https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea_surface_temperature[(2019-01-15T23:00:00Z)][(70):(-10)][(-180):(-100)])



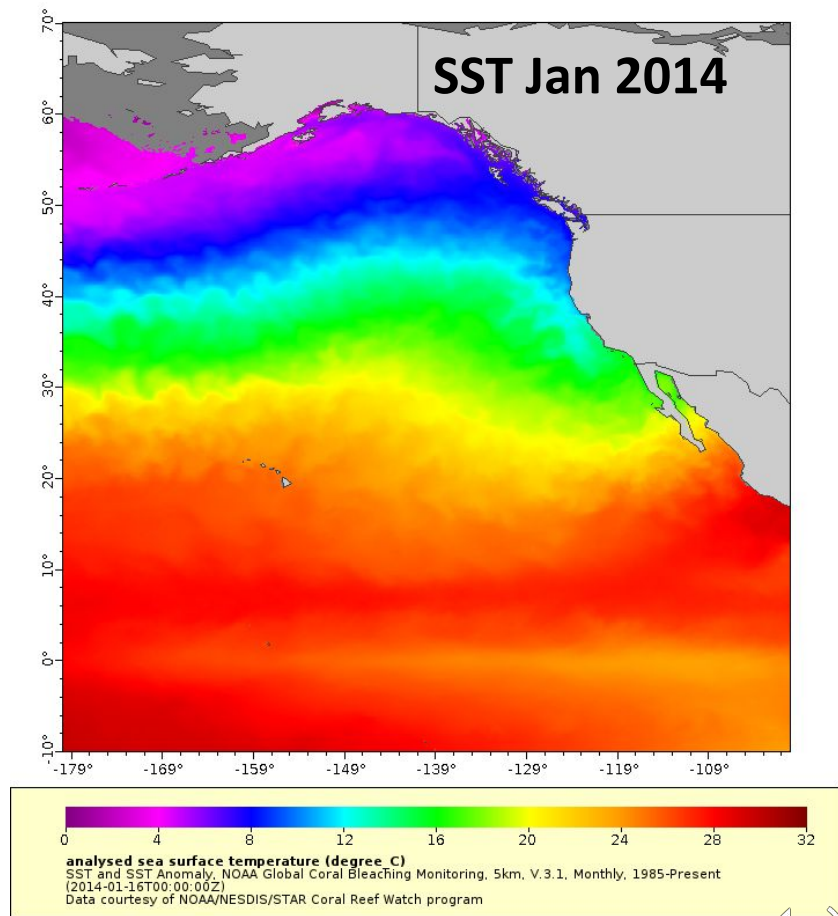
This URL:

[https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea_surface_temperature\[\(2014-01-15\)\]\[\[\(70\):\(-10\)\]\[\[\(-180\):\(-100\)\]\]](https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea_surface_temperature[(2014-01-15)][[(70):(-10)][[(-180):(-100)]])

Produces this figure →

Note:

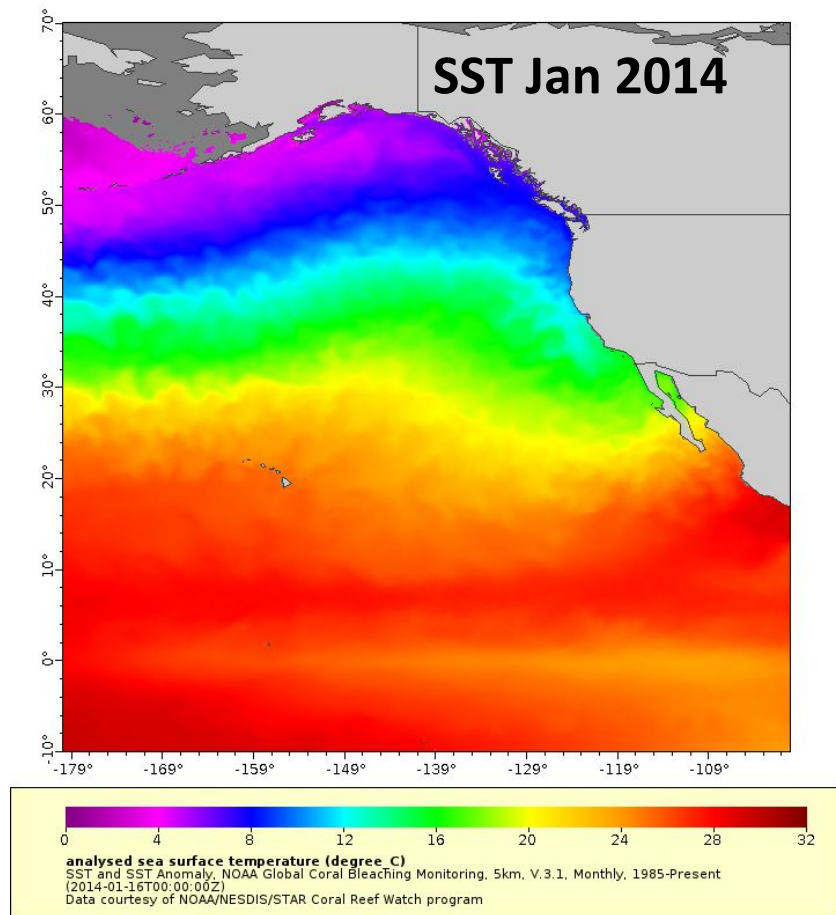
You can download the data in a netCDF file
by changing **.largePng** to **.nc** in the URL



Change the variable:

[https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea_surface_temperature\[\(2019-09-15\)\]\[\[\(70\):\(-10\)\]\[\(-180\):\(-100\)\]\]](https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea_surface_temperature[(2019-09-15)][[(70):(-10)][(-180):(-100)]])

- Change the variable displayed to see the SST anomaly
- For this dataset we will change it to sea_surface_temperature_anomaly

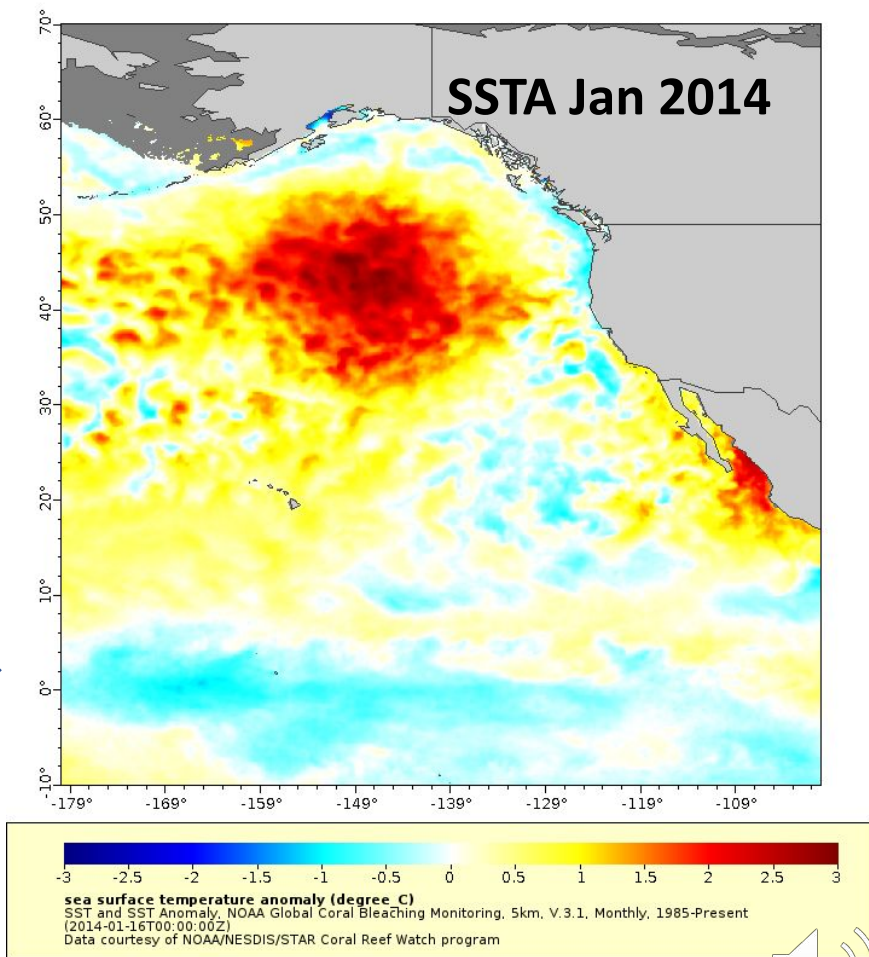


Visualize the Pacific marine heat wave:

[https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea_surface_temperature_anomaly\[\(2014-01-15\)\]\[\(70\):\(-10\)\]\[\(-180\):\(-100\)\]](https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea_surface_temperature_anomaly[(2014-01-15)][(70):(-10)][(-180):(-100)])

Produces this figure →

Note: Changing the variable name produces an anomaly because this dataset has a variable with the SST anomaly in it. Most datasets do not have an anomaly variable in them, so this modification will only work for this dataset.



Create a 2D timeseries:

[https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea_surface_temperature_anomaly\[\(2014-01-15\)\]\[\(70\):\(-10\)\]\[\(-180\):\(-100\)\]](https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea_surface_temperature_anomaly[(2014-01-15)][(70):(-10)][(-180):(-100)])

Next we will examine the temporal evolution of the warm “blob” by making a Hovmöller diagram, a hybrid map with time on one axis, and latitude or longitude on the other. We will make a slice through 45°N.

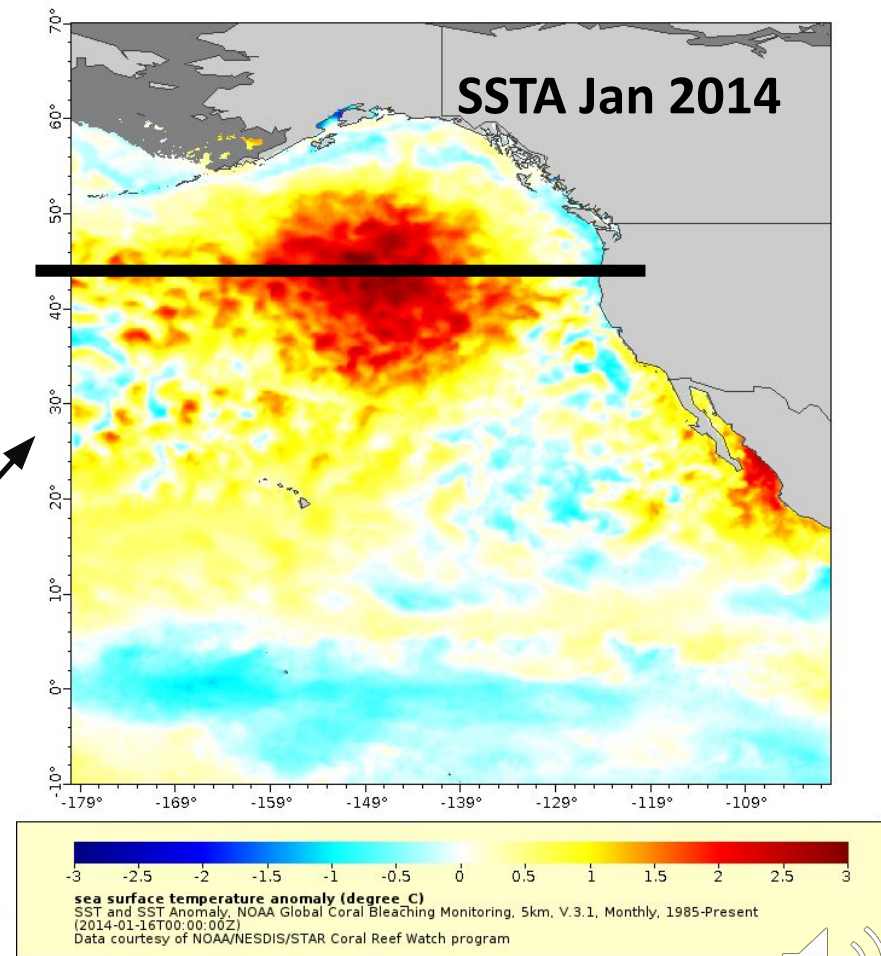
We can do this by setting the y-axis to time on the “Make a Graph” page:

Graph Type: surface

X Axis: longitude

Y Axis: time

Color: sea_surface_temperature_anomaly

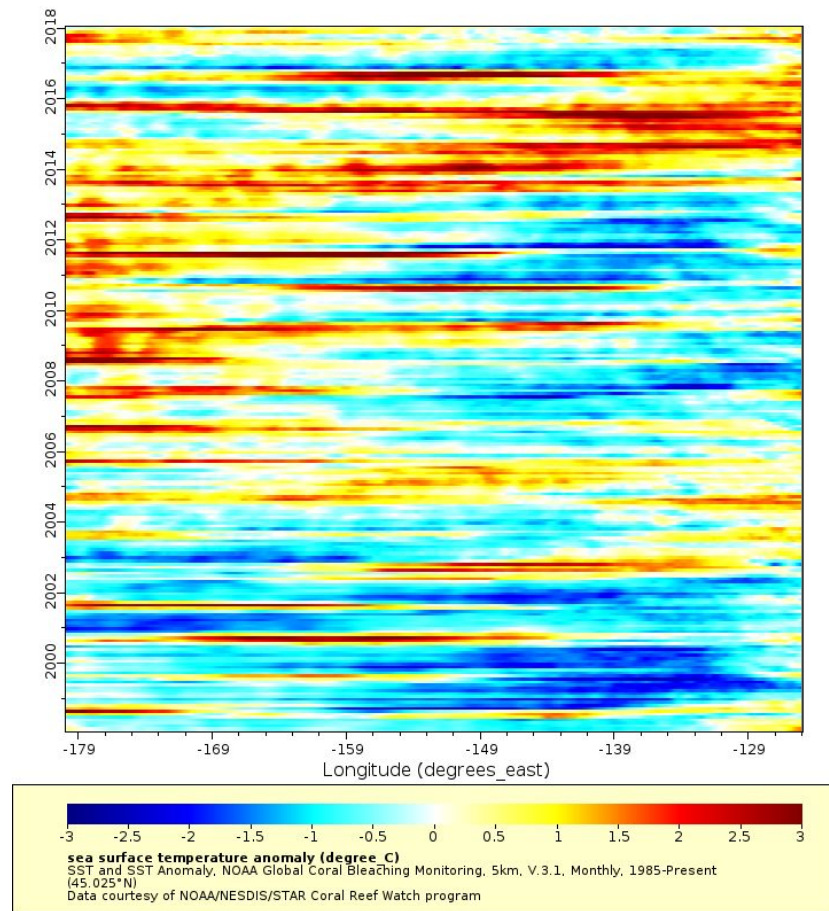


Generate a Hovmöller diagram

[https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea_surface_temperature_anomaly\[\(1985-01-15\):\(2019-12-16\)\]\[\(45\)\]\[\(-180\):\(-125\)\]](https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea_surface_temperature_anomaly[(1985-01-15):(2019-12-16)][(45)][(-180):(-125)])

Produces this figure →

While most of the last 20 years the N. Pacific (at 45°N) has experienced warmer than usual temperatures, only in the past few years has this phenomena spread to coast.

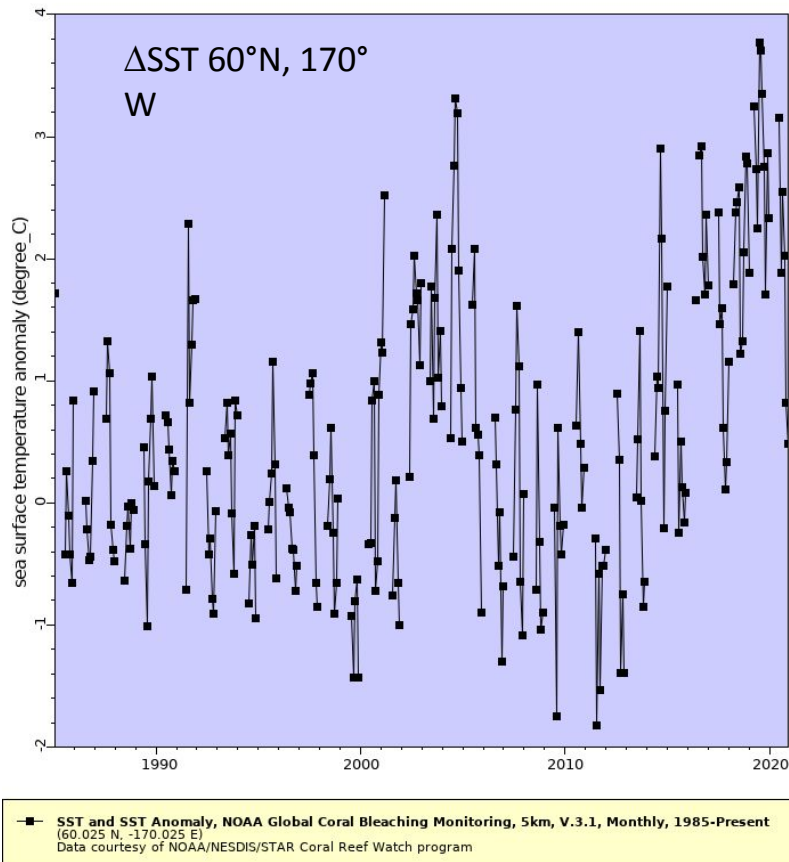


Generate a Timeseries

[https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea_surface_temperature_anomaly\[\(1985-01-16T00:00:00Z\):\(2020-12-16T00:00:00Z\)\]\[\[\(60\)\]\[\(-170\)\]\]](https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea_surface_temperature_anomaly[(1985-01-16T00:00:00Z):(2020-12-16T00:00:00Z)][[(60)][(-170)]])

Produces this figure 

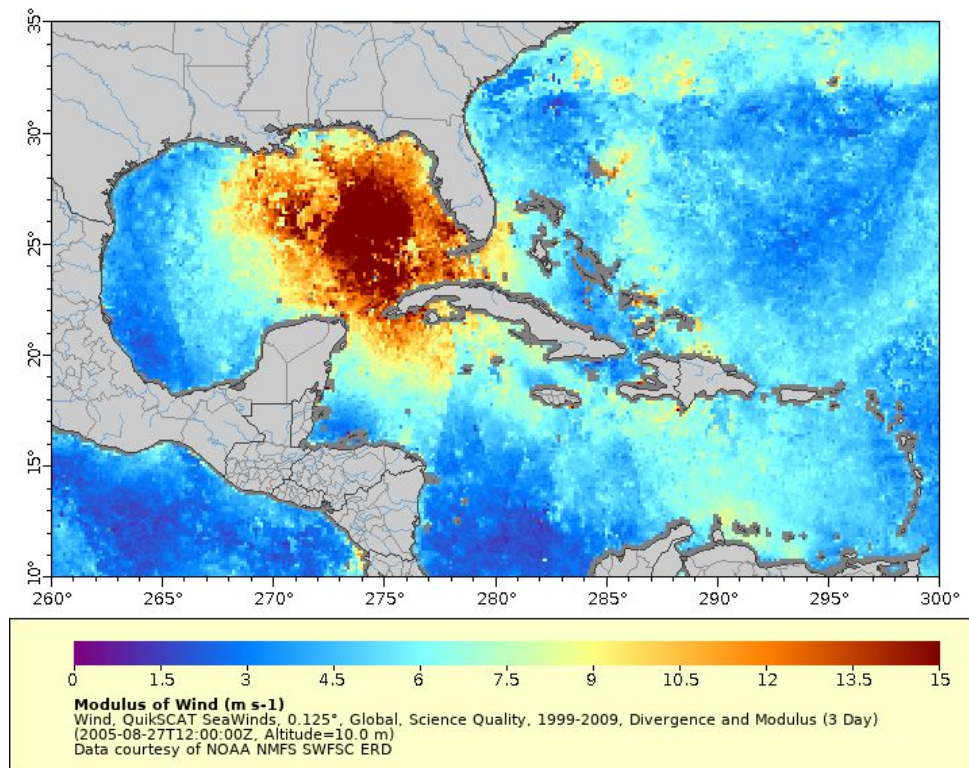
Select 'linesAndMarkers' under Graph Type on the Make a Graph page (.graph) to create a timeseries at any point in the dataset



Visualize wind speeds produced by Hurricane Katrina

[https://coastwatch.pfeg.noaa.gov/erddap/griddap/erdQSdivmod3day.largePng?mod\[\(2005-08-27\)\]\[\(10\)\]\[\(10\):\(35\)\]\[\(260\):\(300\)\]](https://coastwatch.pfeg.noaa.gov/erddap/griddap/erdQSdivmod3day.largePng?mod[(2005-08-27)][(10)][(10):(35)][(260):(300)])

Produces this figure →



Visualize wind vectors produced by Hurricane Katrina



ERDDAP

Easier access to scientific data

ERDDAP > griddap > Make A Graph ?

Dataset Title: **Wind, QuikSCAT SeaWinds, 0.125°, Global, Science Quality, 1999-2009 (3 Day)**

Institution: NOAA NMFS SWFSC ERD (Dataset ID: erdQSwind3day)

Information: [Summary](#) ? | [License](#) ? | [FGDC](#) | [ISO 19115](#) | [Metadata](#) | [Background](#) ? | [Data Access Form](#) | [Files](#)

Graph Type: ?

X Axis: ?

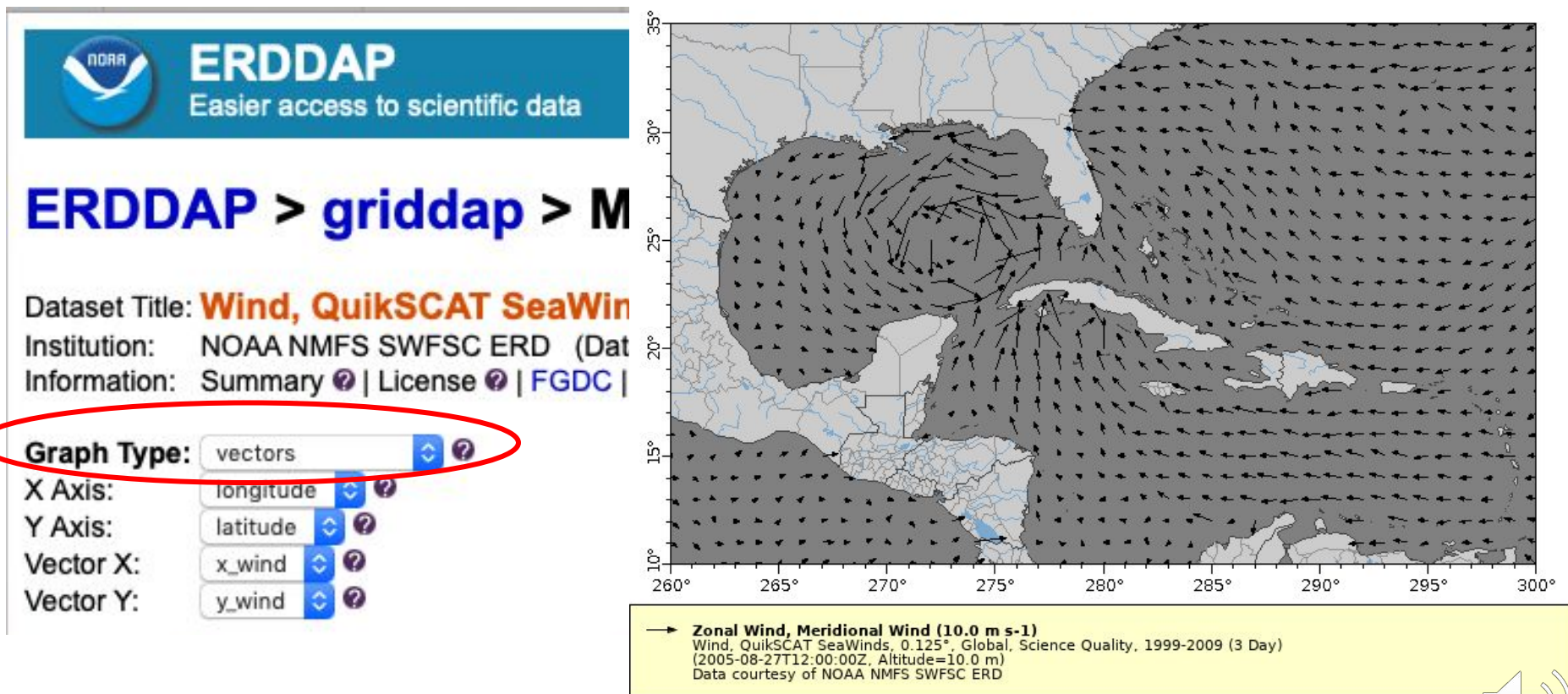
Y Axis: ?

Vector X: ?

Vector Y: ?



Visualize wind vectors produced by Hurricane Katrina



Access tabular data like BGC-Argo Float data

Map of all BGC-Argo floats since 2017-01-01 in the Southern Ocean around South America. Float profiles are colored by date.

https://polarwatch.noaa.gov/erddap/tabledap/SOCCOM_BGC_Argo.graph

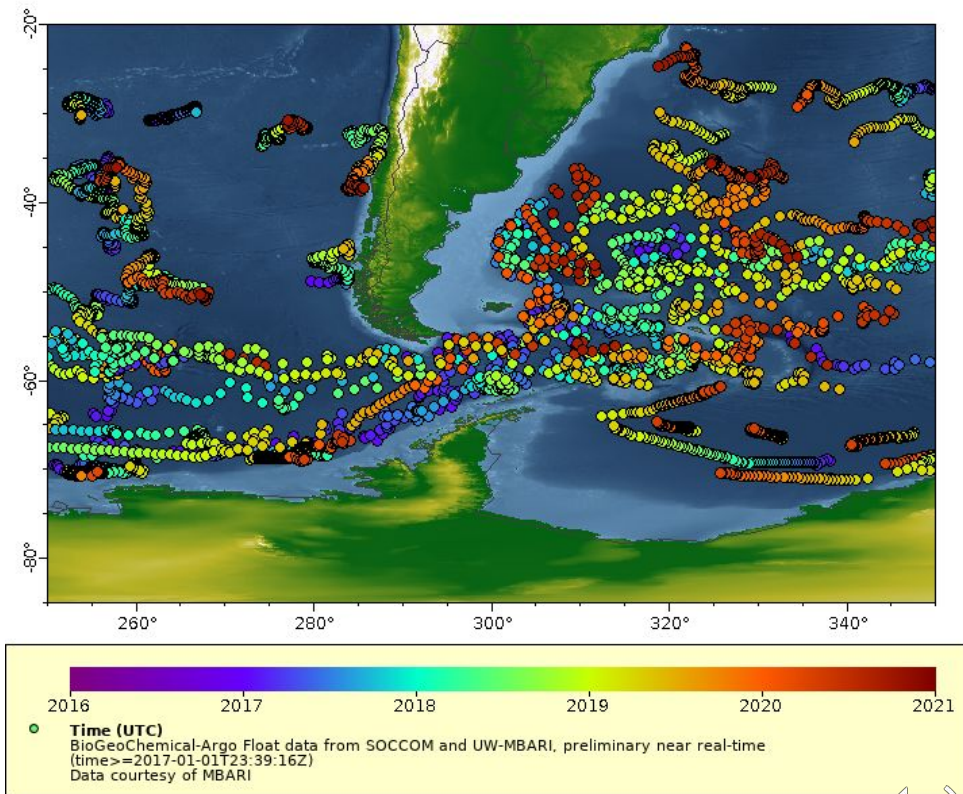
Graph Type:

X Axis:

Y Axis:

Color:

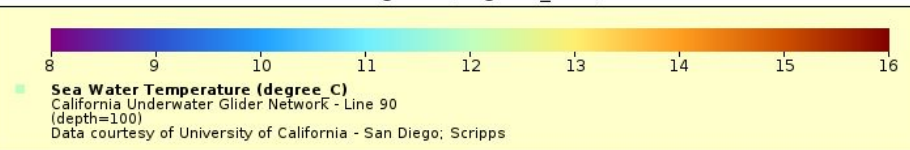
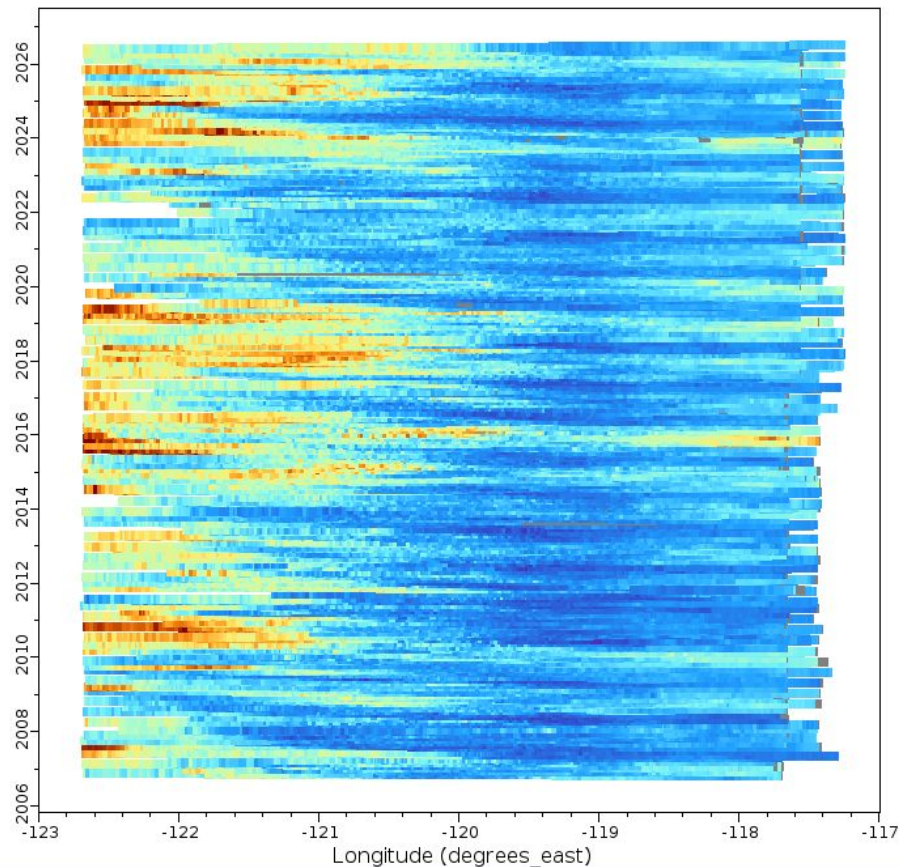
Constraints		Optional Constraint #1	Optional Constraint #2
time	<input <="" td="" type="text" value=">="/> <td> 2017-01-01T23:39:16Z </td> <td><input <="" td="" type="text" value="<="/></td>	2017-01-01T23:39:16Z	<input <="" td="" type="text" value="<="/>
latitude	<input <="" td="" type="text" value=">="/> <td> -85 </td> <td><input <="" td="" type="text" value="<"/></td>	-85	<input <="" td="" type="text" value="<"/>
longitude	<input <="" td="" type="text" value=">="/> <td> 250 </td> <td><input <="" td="" type="text" value="<="/></td>	250	<input <="" td="" type="text" value="<="/>
	<input <="" td="" type="text" value=">="/> <td> </td> <td><input <="" td="" type="text" value="<="/></td>		<input <="" td="" type="text" value="<="/>
	<input <="" td="" type="text" value=">="/> <td> </td> <td><input <="" td="" type="text" value="<="/></td>		<input <="" td="" type="text" value="<="/>



Visualizations of tabular data

<https://spraydata.ucsd.edu/erddap/taledap/binnedCUGN90.graph>

- Hovmoller of temperature along line 90 at 100 m depth from 2006-2026
- This is postprocessed, delayed mode data that has been QCed. Outliers have been removed and data binned at 10m vertical resolution for ease of use. This dataset contains data from multiple glider deployments over time at the location identified by the project and site metadata attributes.
- Produced as part of the California Underwater Glider Network (CUGN)
- Bullets 2 & 3 are information found in the dataset's metadata (on the *.index page)



Downloading Data

NOAA_DHW_monthly.largePng?sea_surface_temperature[(2019-09-21T12:00:00Z)]

Example of a URL data request

Base URL: <https://coastwatch.pfeg.noaa.gov/erddap/griddap/>

Dataset ID: NOAA_DHW_monthly

File Type: .largePng (.nc, .mat, .json, .geotif, .kml, .csv...)

Data Request Begins ?

Variable: sea_surface_temperature

Time range: [(2019-09-15T12:00:00Z):(2019-09-15T12:00:00Z)]

Latitude Range: [(70):(-10)]

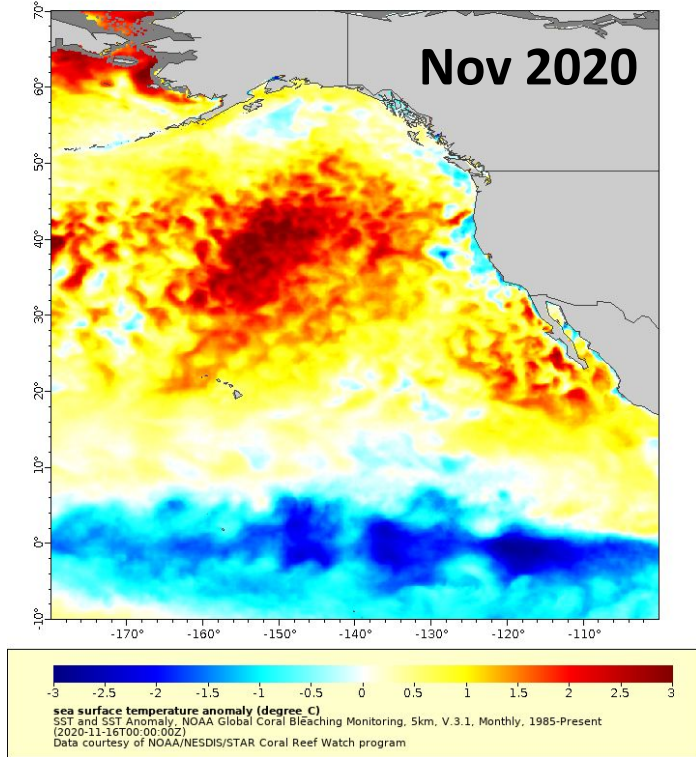
Longitude Range: [(-180):(-100)]

[(70):(-10)][(-180):(-100)]

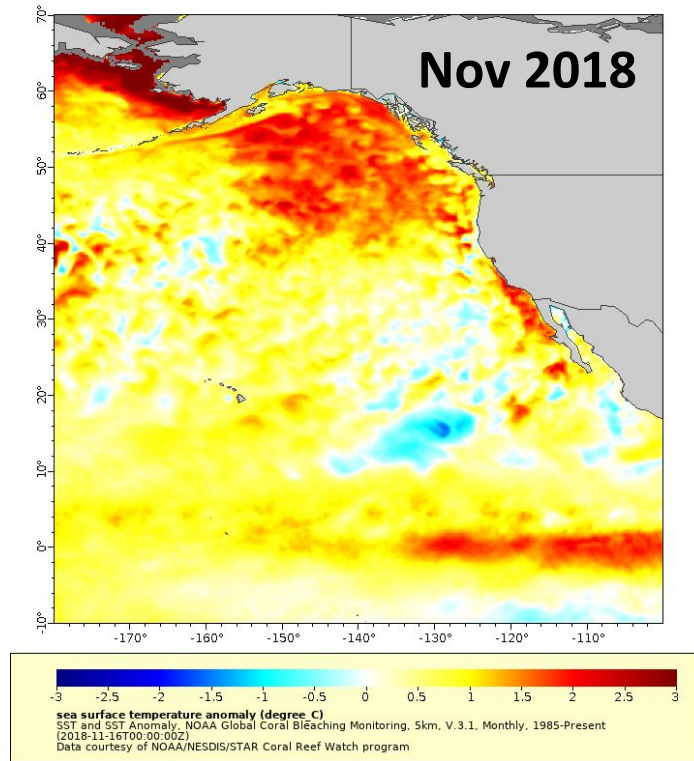


"Last" Data

[https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea surface temperature anomaly\[last\]\[\[\(70\):\(-10\)\]\[\[\(-180\):\(-100\)\]\]](https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea%20surface%20temperature%20anomaly[last][[(70):(-10)][[(-180):(-100)]])



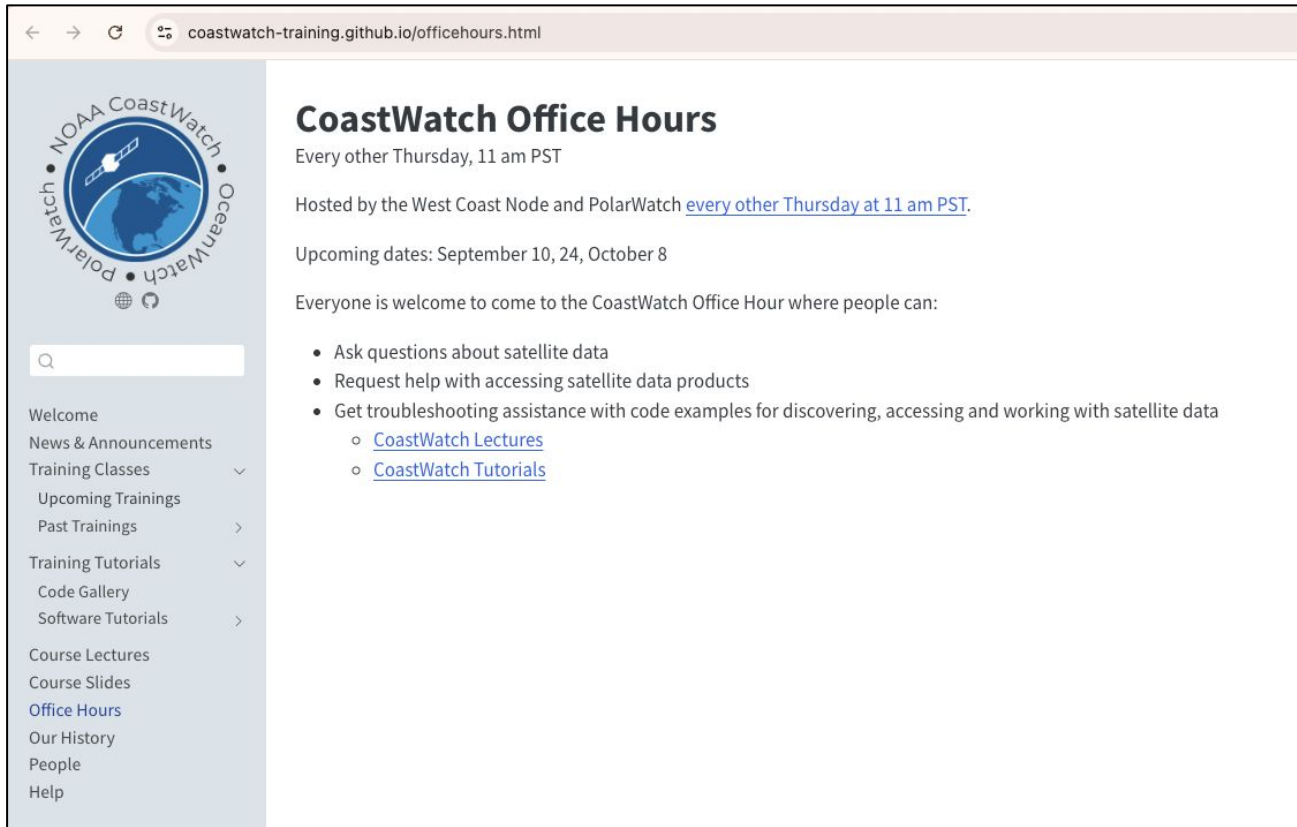
[https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea surface temperature anomaly\[last-24\]\[\[\(70\):\(-10\)\]\[\[\(-180\):\(-100\)\]\]](https://coastwatch.pfeg.noaa.gov/erddap/griddap/NOAA_DHW_monthly.largePng?sea%20surface%20temperature%20anomaly[last-24][[(70):(-10)][[(-180):(-100)]])



Questions?

coastwatch.info@noaa.gov

cara.wilson@noaa.gov



The screenshot shows a web browser window with the address bar displaying `coastwatch-training.github.io/officehours.html`. The page features a sidebar on the left with a search bar and a navigation menu. The main content area is titled "CoastWatch Office Hours" and provides details about the event, including its frequency, host, upcoming dates, and a list of topics for discussion.

CoastWatch Office Hours
Every other Thursday, 11 am PST

Hosted by the West Coast Node and PolarWatch [every other Thursday at 11 am PST](#).

Upcoming dates: September 10, 24, October 8

Everyone is welcome to come to the CoastWatch Office Hour where people can:

- Ask questions about satellite data
- Request help with accessing satellite data products
- Get troubleshooting assistance with code examples for discovering, accessing and working with satellite data
 - [CoastWatch Lectures](#)
 - [CoastWatch Tutorials](#)

Sidebar Navigation:

- Welcome
- News & Announcements
- Training Classes
 - Upcoming Trainings
 - Past Trainings
- Training Tutorials
 - Code Gallery
 - Software Tutorials
- Course Lectures
- Course Slides
- Office Hours
- Our History
- People
- Help

